CHAPTER 3

# Vector calculus and the scalar product

**Key point · Chapter path. Objectives.** Choose between projection, coordinates and angle; calculate a length; characterise orthogonality and a locus.

**Prerequisites.** Vectors, Chasles's relation, Pythagoras, cosine in a right triangle, orthonormal frame.

**Yearly reference point.** 6–15 October: 8 hours

## Diagnostic: six questions, 10 minutes

Answer without consulting the lesson. Mark the questions for which help is needed. This diagnostic is not graded.

1. For $A(1,2)$ and $B(4,-2)$, calculate $\overrightarrow{AB}$.
2. Calculate the norm of $(3,-4)$.
3. Simplify $\overrightarrow{AB}+\overrightarrow{BC}$.
4. Expand $(a+b)^2$.
5. In a right triangle, an angle has adjacent side 3 and hypotenuse 5. Give its cosine.
6. What is the midpoint of $A(-2,1)$ and $B(4,3)$?

**After the diagnostic.** Revisit the flagged topics with the teacher; redo a similar question, then explain the rule used.

**Speaking and logic.** Compare, on the same problem, coordinates and vector calculus without a frame. Justify why your method uses the data most directly.

## Investigate before formalising

Construct three points M such that $\overrightarrow{MA} \cdot \overrightarrow{MB} = 0$ for a segment AB of 6 cm. Conjecture the locus, then produce a proof that includes A and B.

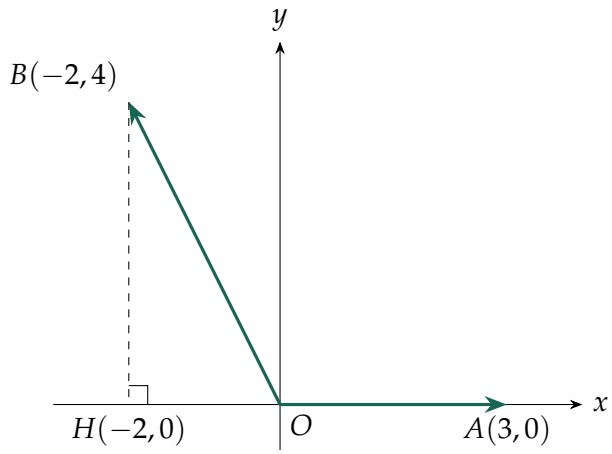
**Organisation.** Individual investigation 5 min, comparison in pairs 5 min, class discussion 5 min. Keep a record of one attempt, even an unfinished one. A conjecture must then be justified.

## 3.1 Projection and angle

**Key point · Definition and results.** For two nonzero vectors $\vec{u}, \vec{v}$, their scalar product is the real number $\vec{u} \cdot \vec{v} = \|\vec{u}\| \, \|\vec{v}\| \cos\theta$, where $\theta \in [0, \pi]$ is their unoriented angle. With a zero vector, the product is 0. If $H$ is the projection of B onto $(OA)$, $\overrightarrow{OA} \cdot \overrightarrow{OB}$ equals $OA \times OH$ when H is on the ray $[OA)$, and its opposite otherwise.

**Status and justification.** Definition; the equivalence with the projection follows from the cosine in right triangles, with a negative algebraic projection for an obtuse angle.

**Example 1.** For two vectors of norms 3 and 4 forming 60°, the product is $\dfrac{3 \times 4 \times 1}{2} = 6$. With 120°, it is $-6$. If $A(3,0)$ and $B(-2,4)$, the projection is $H(-2,0)$ and the product is $-6$.




The projection H lies on the negative side of the horizontal axis. Hence $\overrightarrow{OA} \cdot \overrightarrow{OB} = 3 \times (-2) = -6$.

**Watch out.** The result is a number, not a vector. A right angle corresponds to 0, not to 1.

**Application.** Two vectors have norms 2 and 5 and form 90°, then 180°. Calculate the products.

## 3.2 Coordinates, norm and orthogonality

**Key point · Definition and results.** In an orthonormal basis, for $\vec{u} = (x, y)$ and $\vec{v} = (x', y')$, $\vec{u} \cdot \vec{v} = xx' + yy'$ and $\|\vec{u}\|^2 = x^2 + y^2$. The vectors are orthogonal if and only if their product is zero. For an orthonormal basis $(\vec{i}, \vec{j})$, $x = \vec{u} \cdot \vec{i}$ and $y = \vec{u} \cdot \vec{j}$.

**Status and justification.** Properties proved using bilinearity: expand $(x\vec{i} + y\vec{j}) \cdot (x'\vec{i} + y'\vec{j})$ and use $\vec{i} \cdot \vec{i} = \vec{j} \cdot \vec{j} = 1$, $\vec{i} \cdot \vec{j} = 0$.

**Example 2.** $\vec{u} = (2,3)$, $\vec{v} = (3,-2)$: product $6 - 6 = 0$. The two vectors are orthogonal. Their norm is $\sqrt{13}$. The product of $(2,3)$ with $(1,0)$ is 2: we recover the x-coordinate.

**Watch out.** The usual formula assumes an orthonormal basis. Collinearity and orthogonality are different properties.

**Application.** For which real number t are the vectors $(t, 2)$ and $(3, -6)$ orthogonal?

## 3.3 Vector calculus and identities

**Key point · Definition and results.** The scalar product is symmetric and bilinear: $\vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}$ and $(\lambda\vec{u}) \cdot \vec{v} = \lambda(\vec{u} \cdot \vec{v})$. Hence $\|\vec{u} \pm \vec{v}\|^2 = \|\vec{u}\|^2 \pm 2\vec{u} \cdot \vec{v} + \|\vec{v}\|^2$. It follows that $\vec{u} \cdot \vec{v} = \dfrac{\|\vec{u} + \vec{v}\|^2 - \|\vec{u}\|^2 - \|\vec{v}\|^2}{2}$.

**Status and justification.** Bilinearity accepted from the geometric definition; symmetry comes from the cosine. The identities are proved by expanding $(\vec{u} \pm \vec{v}) \cdot (\vec{u} \pm \vec{v})$.

**Example 3.** If $\|\vec{u}\| = 3$, $\|\vec{v}\| = 4$ and $\|\vec{u} + \vec{v}\| = 5$, the product is $\dfrac{25 - 9 - 16}{2} = 0$. We deduce orthogonality without coordinates.

**Watch out.** In general, $\|\vec{u} + \vec{v}\| \neq \|\vec{u}\| + \|\vec{v}\|$. The identities concern the squares of the norms.

**Application.** Given $\|\vec{u}\| = 2$, $\|\vec{v}\| = 3$, $\vec{u} \cdot \vec{v} = 1$. Calculate $\|\vec{u} - \vec{v}\|$.

## 3.4 Al-Kashi and choosing a method

**Key point · Definition and results.** In a triangle ABC, $BC^2 = AB^2 + AC^2 - 2AB\,AC\cos\widehat{BAC}$. This formula allows us to calculate the third side from two sides and the included angle, or to find the cosine of an angle from the three sides.

**Status and justification.** By Chasles, $\overrightarrow{BC} = \overrightarrow{AC} - \overrightarrow{AB}$. Expanding its scalar square gives $BC^2 = AC^2 + AB^2 - 2\overrightarrow{AB} \cdot \overrightarrow{AC}$; then introduce the expression with the cosine.

**Example 4.** $AB = 5$, $AC = 3$ and $\widehat{BAC} = 60°$: $BC^2 = 25 + 9 - 15 = 19$, so $BC = \sqrt{19}$. With three sides, isolate the cosine before approximating the angle on a calculator, if one is allowed.

**Watch out.** The angle is included between the two sides of the product term. The minus sign is essential.

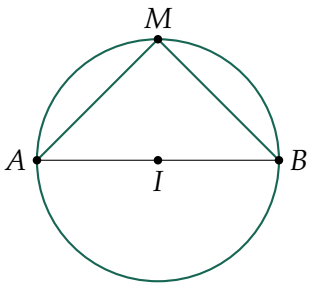
**Application.** A triangle has $AB = AC = 4$ and $BC = 4\sqrt{2}$. Determine the angle A.

## 3.5 The circle with diameter AB

**Key point · Definition and results.** For two distinct points A and B with midpoint I, $\overrightarrow{MA} \cdot \overrightarrow{MB} = MI^2 - \dfrac{AB^2}{4}$. The set of points M such that $\overrightarrow{MA} \cdot \overrightarrow{MB} = 0$ is the circle with centre I and radius $\dfrac{AB}{2}$, including the points A and B.

**Status and justification.** Write $\overrightarrow{MA} = \overrightarrow{MI} + \overrightarrow{IA}$ and $\overrightarrow{MB} = \overrightarrow{MI} - \overrightarrow{IA}$. The cross terms cancel; what remains is $MI^2 - IA^2$. Since $IA = \dfrac{AB}{2}$, equality to zero is equivalent to the metric equation of the circle.

**Example 5.** $A(-2,0)$ and $B(2,0)$: I is the origin. The locus is $x^2 + y^2 = 4$. For $M(0,1)$, the product is $1 - 4 = -3$; for $M(0,2)$, it is 0.




Circle with diameter AB: the vectors $\overrightarrow{MA}$ and $\overrightarrow{MB}$ are orthogonal.

**Watch out.** The endpoints belong to the locus even though "right triangle" does not apply to a degenerate triangle.

**Application.** For $A(0,0)$, $B(6,0)$, determine the points M such that $\overrightarrow{MA} \cdot \overrightarrow{MB} = 7$.

## Exercise path

Start with exercises 1 and 2. Then choose one of exercises 3 or 4 and do it without help. Exercises 5 to 7 are a reserve for transfer, investigation or programming: the teacher designates one. Exercise 8 is revisited after a delay.

### Exercise 3.1 · Direct calculations

In an orthonormal basis of the plane, consider the vectors $\vec{u} = (1, -2)$ and $\vec{v} = (4, 3)$.

1. Calculate the scalar product $\vec{u} \cdot \vec{v}$, then the norms $\|\vec{u}\|$ and $\|\vec{v}\|$.
2. Calculate the coordinates of $\vec{u} + \vec{v}$ and deduce $\|\vec{u} + \vec{v}\|^2$.
3. Recover this last result using the identity involving $\vec{u} \cdot \vec{v}$.

### Exercise 3.2 · A right triangle?

The plane is equipped with an orthonormal frame. Given the points $A(1, 1)$, $B(4, 2)$ and $C(0, 4)$.

1. Calculate the coordinates of $\overrightarrow{AB}$ and $\overrightarrow{AC}$, then their scalar product.
2. Deduce that the triangle $ABC$ is right-angled; state at which vertex.
3. Calculate the lengths of the two perpendicular sides, then the area of the triangle in area units.

### Exercise 3.3 · Length without a frame

The quadrilateral $ABCD$ is a parallelogram with $AB = 4$, $AD = 3$ and $\widehat{BAD} = 60°$. The lengths are expressed in the same unit.

1. Draw a figure and calculate the scalar product $\overrightarrow{AB} \cdot \overrightarrow{AD}$.
2. Determine the exact lengths of the diagonals $[AC]$ and $[BD]$. Explain how the vectors corresponding to the diagonals are expressed in terms of $\overrightarrow{AB}$ and $\overrightarrow{AD}$.
3. State which diagonal is longer and check that this result is consistent with your figure.

### Exercise 3.4 · Recovering an angle

Two nonzero vectors $\vec{u}$ and $\vec{v}$ satisfy $\|\vec{u}\| = 2$, $\|\vec{v}\| = 6$ and $\vec{u} \cdot \vec{v} = -6$. Let $\theta$ be their unoriented angle, between $0°$ and $180°$.

1. Determine the exact value of $\cos\theta$, then the measure of $\theta$.
2. Explain how the sign of the scalar product made it possible to predict that the angle would be obtuse.

### Exercise 3.5 · A variable locus

The points $A$ and $B$ are distinct and fixed. Let $I$ be the midpoint of $[AB]$. For a fixed real number $k$, we look for the set of points $M$ of the plane that satisfy $\overrightarrow{MA} \cdot \overrightarrow{MB} = k$.

1. Express this scalar product in terms of $MI$ and $AB$ using Chasles's relation and a scalar identity.
2. Describe precisely the set sought according to the value of $k$. For each case, state whether it is empty, a single point or a circle; then give its characteristics.
3. Explain the role of the fact that a squared distance is positive or zero.

### Exercise 3.6 · Work done by a force

An object moves in a straight line from $A$ to $B$ over a distance of 4 m. It is subject to a constant force $\vec{F}$ of magnitude 50 N. The work done by this force is defined by $W = \vec{F} \cdot \overrightarrow{AB}$ and is expressed in joules.

1. Calculate $W$ when the angle between $\vec{F}$ and $\overrightarrow{AB}$ is $60°$.
2. Repeat the calculation for an angle of $120°$.
3. Compare the signs of the two works and explain, in each situation, whether the force helps the displacement or opposes it.

### Exercise 3.7 · Contrapositive

We wish to prove the following statement, valid for two vectors of the plane:

$$\|\vec{u}+\vec{v}\|^2 \neq \|\vec{u}\|^2 + \|\vec{v}\|^2 \quad \Longrightarrow \quad \vec{u} \text{ and } \vec{v} \text{ are not orthogonal.}$$

1. State the contrapositive of this implication.
2. Prove this contrapositive using a scalar identity, then conclude about the initial statement.

### Exercise 3.8 · Delayed review

A triangle $ABC$ satisfies $AB = 3$, $AC = 4$ and $BC = 5$. We look for $\overrightarrow{AB} \cdot \overrightarrow{AC}$ without using coordinates.

1. Calculate this scalar product using a formula involving the three lengths.
2. Recover the result by determining the nature of the triangle. Name the theorem used and its numerical hypothesis.

**Homework.** Do only the task designated at the end of the session, in the dated progression: 10–20 min at most. The exercises depend on the topics already studied, so some questions are deferred. Revisit exercise 8 one week after the chapter (10 min). If the allotted time is exceeded, note the difficulty and bring your attempts; do not pile up the whole reserve.

## To do in the CQFD textbook

These exercises are done in your textbook. The progression indicates the session; they replace the homework planned for that day (10 to 20 minutes in total).

**CQFD, exercise 9 p. 250, questions a and b.** Calculate the scalar product in the orthonormal basis and detail the two products of coordinates.

**CQFD, exercise 11 p. 250, questions 1, 2 and 3.** Use in turn the coordinates, Chasles's relation and the fact that I is the midpoint.

## MCQ 1: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. $(2,1)\cdot(3,-4)$ equals
   - A. 2
   - B. 10
   - C. $(6,-4)$
   - D. $-2$
2. The norm of $(-3,4)$ equals
   - A. 7
   - B. 25
   - C. 5
   - D. 1
3. Two vectors with zero product are
   - A. equal
   - B. collinear
   - C. opposite
   - D. orthogonal
4. Norms 2 and 3, angle $60^\circ$: product
   - A. 6
   - B. 3
   - C. $\sqrt{3}$
   - D. 0
5. $\|u-v\|^2$ equals
   - A. $\|u\|^2-\|v\|^2$
   - B. $\|u\|^2+\|v\|^2$
   - C. $\|u\|^2-2u\cdot v+\|v\|^2$
   - D. $\|u\|^2+2u\cdot v+\|v\|^2$

## MCQ 2: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. An obtuse angle between nonzero vectors gives a product that is
   A. positive
   B. zero
   C. equal to 1
   D. negative
2. In an orthonormal basis, $u \cdot i$ is
   A. the x-coordinate of u
   B. the norm of u
   C. the y-coordinate of u
   D. always zero
3. The locus $\overrightarrow{MA} \cdot \overrightarrow{MB} = 0$ for $A \neq B$ is
   A. the perpendicular bisector
   B. the line AB
   C. the circle with diameter AB
   D. the midpoint alone
4. If $u \cdot v = 3$, $(2u) \cdot (-v)$ equals
   A. $-3$
   B. $-6$
   C. $6$
   D. $9$
5. For $AB = AC = 2$ and a right angle at A, $BC$ equals
   A. $4$
   B. $2$
   C. $8$
   D. $2\sqrt{2}$

### True or false? Justify.

1. The scalar product of two vectors is always positive.
2. The zero vector is orthogonal to every vector.
3. $u \cdot v = 0$ implies that one of the vectors is zero.

> **Key point · Self-assessment.** I distinguish the norm from its square; I choose a suitable formula; I check the frame; I write an equivalence for a locus.
>
> For each objective, indicate: achieved without help; achieved with help; to be reworked. Write one example and one mistake to avoid.

# Solutions and teaching notes for chapter 3

**Teacher use.** The answers below concern the tasks not worked in the lesson. First have students state their method; give graded help before the solution. The durations are estimates to be calibrated in class.

## Diagnostic and remediation

1. $(3, -4)$. *Action:* Subtract end minus start on two drawings.
2. $\sqrt{3^2 + (-4)^2} = 5$. *Action:* Review Pythagoras and distinguish norm from coordinates.
3. $\overrightarrow{AC}$. *Action:* Draw the composite displacement.
4. $a^2 + 2ab + b^2$. *Action:* Identify the cross term.
5. $\frac{3}{5}$. *Action:* Name the sides relative to the angle.
6. $(1, 2)$. *Action:* Average the two coordinates separately.

## Investigation activity

Take points on the circle with diameter AB; proof via the midpoint I: product $MI^2 - 9$. The endpoints are included because one of the vectors is zero.

## Immediate applications

1. 0 then $-10$. The cosine is 0 and $-1$ respectively.
2. $3t - 12 = 0$, so $t = 4$.
3. The square is $4 - 2 + 9 = 11$; the norm is $\sqrt{11}$.
4. $32 = 16 + 16 - 32\cos A$, so $\cos A = 0$ and $A = 90°$.
5. $MI^2 - 9 = 7$: circle with centre $(3, 0)$ and radius 4.

## Exercises: method and results

**Exercise 3.1 – Direct calculations.** Product $4 - 6 = -2$; norms $\sqrt{5}$ and 5; sum $(5, 1)$ with square 26. Check: $5 + 2(-2) + 25 = 26$.

**Exercise 3.2 – A right triangle?** $\overrightarrow{AB} = (3, 1)$, $\overrightarrow{AC} = (-1, 3)$; product 0. $AB = AC = \sqrt{10}$, area $\frac{AB \times AC}{2} = 5$ area units.

**Exercise 3.3 – Length without a frame.** $\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{AD}$ and $\overrightarrow{BD} = \overrightarrow{AD} - \overrightarrow{AB}$. Product $\frac{4 \times 3}{2} = 6$; $AC^2 = 16 + 9 + 12 = 37$ and $BD^2 = 16 + 9 - 12 = 13$.

**Exercise 3.4 – Recovering an angle.** $\cos\theta = -\frac{6}{12} = -\frac{1}{2}$; in $[0, 180°]$, the angle is 120°. The domain ensures uniqueness.

**Exercise 3.5 – A variable locus.** $MI^2 = k + \dfrac{AB^2}{4}$. If the right-hand side is negative: empty set; zero: the single point I; positive: circle whose radius is its square root. The discussion bears on a squared distance.

**Exercise 3.6 – Work done by a force.** $W = \dfrac{50 \times 4 \times 1}{2} = 100$ J. At 120°, $W = -100$ J. The sign distinguishes help to the displacement from opposition; it does not describe a negative distance.

**Exercise 3.7 – Contrapositive.** The direct property "orthogonal implies equal squares" follows from the expansion. Its contrapositive is exactly the statement. We can also write $2\vec{u} \cdot \vec{v} \neq 0$.

**Exercise 3.8 – Delayed review.** Al-Kashi: $\dfrac{9 + 16 - 25}{2} = 0$. Converse of Pythagoras: triangle right-angled at A, so zero product. Name the hypotheses of each method.

**CQFD, exercise 9 p. 250, questions a and b.** a: $1 \times 3 + 2 \times (-1) = 1$. b: $(-1) \times 12 + 3 \times 7 = 9$. The scalar product is a number, not a vector.

**CQFD, exercise 11 p. 250, questions 1, 2 and 3.** $\overrightarrow{AB} = (4,6)$, $\overrightarrow{AC} = (-3,4)$, $I(-1,-2)$ and $\overrightarrow{CI} = (5,-1)$. Hence $\overrightarrow{AB} \cdot \overrightarrow{AC} = 12$, $\overrightarrow{AB} \cdot \overrightarrow{CI} = 14$. Chasles gives $\overrightarrow{AB} \cdot \overrightarrow{AI} = 12 + 14 = 26$. Two checks: $\overrightarrow{AI} = (2,3)$ gives $8 + 18 = 26$; or $\overrightarrow{AI} = \dfrac{1}{2}\overrightarrow{AB}$ gives $\dfrac{1}{2}(16 + 36) = 26$.

## MCQ: answers and reasons

**MCQ 1.**

1. A. $6 - 4 = 2$.
2. C. Square root of 25.
3. D. Scalar criterion.
4. B. $\dfrac{2 \times 3}{2} = 3$.
5. C. Expand the scalar square.

**MCQ 2.**

1. D. Negative cosine.
2. A. Coordinates by projection.
3. C. $MI^2 = \dfrac{AB^2}{4}$.
4. B. Bilinearity: $2(-1)3$.
5. D. $BC^2 = 8$.

### True or false

1. False: $u \cdot (-u) = -\|u\|^2 < 0$ for $u \neq 0$.
2. True: its product with every vector is $0$.
3. False: $(1,0)$ and $(0,1)$ are nonzero and orthogonal.

**Differentiated teaching.** For a technical block, offer only the first step (domain, factor or reference sample space), then withdraw the help. For an independent student, ask for a second method and the treatment of a limiting case. Orally, allow an initial reformulation in the language the student has mastered, then require a precise mathematical sentence in French.